From Single-Species to Ecosystem-Informed Fisheries Advice: Stress-Testing the Northeast Atlantic Mackerel Management Framework

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Abstract

Small to intermediate-sized pelagic fish such as Northeast Atlantic mackerel support valuable commercial fisheries and play vital roles in marine ecosystems. However, single-species advice often fails to consider trade-offs between social, economic and ecological objectives, including trophic requirements of predators, environmental effects and climate change. Here, we show how ecosystem considerations can be incorporated into single-species advice through a retrospective Management Strategy Evaluation (MSE) that backtests historical mackerel advice over 2005–2020. We stress-test the ICES advice rule as a closed-loop management procedure under epistemic uncertainty, which is potentially explainable or reducible, and aleatory uncertainty, which is random and irreducible. Scenarios include variation in predation mortality informed by strategic ecosystem models, while the estimation model retains constant natural mortality as in the ICES assessment. We evaluate potential Ecological Reference Points that consider forage availability for predators, age structure, and rebuilding time alongside traditional performance metrics related to maximum sustainable yield. Results showed that current advice frameworks performed well under historical assumptions but failed to meet ecological objectives when environmental conditions shift. It was shown that integrating ecosystem information into advice systems is possible through incremental, pragmatic steps without undermining existing single-species structures. Ecosystem-informed MSE provides a credible tool for evaluating trade-offs, guiding adaptive management, and aligning fisheries advice with sustainability objectives.

Introduction

Small to intermediate-sized pelagic fish, such as sprat (*Sprattus sprattus*), herring (*Clupea harengus*), and Northeast Atlantic mackerel (*Scomber scombrus*), occupy a critical position in marine ecosystems acting as hub species (Fulton and Sainsbury. 2024) by transferring energy from lower trophic levels (e.g., zooplankton) to higher trophic levels including seabirds, marine mammals, and piscivorous fish (e.g., Carroll et al. 2017; Dickey-Collas et al. 2014; Trenkel et al. 2014; Gjøsæter, Bogstad, and Tjelmeland 2009; Fulton and Sainsbury 2024; Ruzicka et al. 2024). These species are often termed “forage fish”, generally considered to be small to intermediate-sized species that form large schools or aggregations, are generally short-lived, and exhibit significant natural fluctuations in population size driven by environmental conditions and anthropogenic factors (Smith et al. 2011). The term small to intermediate-sized pelagic emphasises their open-water habitat and schooling behaviour; the term forage fish highlights their role as prey for higher trophic levels. For simplicity, we use the term forage fish throughout this paper.

As well as a vital energy pathway for marine food web structure, forage fish maintain the resilience of ocean food webs by enabling them to bounce back from disruptions and maintain their essential functions while also supporting valuable commercial fisheries, making them both ecologically and economically significant (Pikitch et al. 2014; Margaret C. Siple, Essington, and E. Plagányi 2019). Given their widespread distribution, fisheries targeting forage fish are often shared between nation states and may occur in Areas Beyond National Jurisdiction (ABNJ). Climate-driven shifts in distribution are likely to intensify these shared-stock and governance challenges, creating or exacerbating conflicts over access, allocation, and responsibility for management (Gullestad, Sundby, and Kjesbu 2020). The management of these species presents unique challenges due to their susceptibility to environmental stressors, climate variability, and predator-prey dynamics that can significantly impact their productivity and population structure (Rooper et al. 2024). Traditional single-species assessment approaches, particularly for pelagic fish, are increasingly incorporating ecosystem trends and variability, however this inclusion is almost always implicit: the assessments do not explicitly consider the complex ecological interactions and environmental dependencies that characterise small pelagic fish populations (Trenkel et al. 2023). Nor do they explicitly consider the needs of predators (ICES 2023a), or how these may influence progress towards meeting multiple policy objectives, such as concurrently achieving commitments for sustainable exploitation (e.g., Maximum Sustainable Yield (MSY)) and ecosystem health (e.g., Good Environmental Status (GES)).

Current Management Frameworks and Limitations

Efforts to achieve ecosystem objectives are currently being applied via frameworks such as the Ecosystem Approach to Fisheries Management (EAFM) and Ecosystem-Based Fisheries Management (EBFM) (Pikitch et al. 2004; Link, Griffis, and Busch 2020). Conventionally, EAFM involves considering ecosystem information alongside traditional single-species advice, while EBFM extends management strategies to incorporate ecosystem information for managing multiple species across the ecosystem (Link 2010; Link and Browman 2014;

Howell et al. 2021; Dickey-Collas et al. 2022; Muffley et al. 2021). Here, we propose that it is more useful, both for recognising general progress and encouraging wider implementation by decision makers, to consider EBFM as the primary terminology and to view it as a spectrum of incremental management steps, moving from ecosystem-informed single species management as the initial stage, towards more integrated forms of multispecies management. This paper therefore uses the term EBFM throughout. A crucial aspect of EBFM are the three pillars of sustainable development (Karnauskas et al. 2021), which involves considering trade-offs between social, economic, and ecological objectives. In ICES (International Council for the Exploration of the Sea) , ecosystem information has been explicitly integrated into single-species advice through pragmatic solutions such as dynamic harvest control rules, e.g. F_{eco} an operational reference point that enhances existing single species advice, for example by decreasing the total allowable catch (TAC) when stock productivity is reduced due to adverse environmental conditions (Howell et al. 2021; Bentley et al. 2021).

Conventional single-species fisheries management advice is mainly based on stock biomass and exploitation levels, quantities that are not directly observable and must be estimated using stock assessment models (Hilborn and Walters 2013). However, recent studies have shown that estimates of stock status and trends may be biased (Edgar et al. 2024), exacerbated by the fact that the models used to generate them are rarely validated against observations (Kell et al. 2024). This lack of validation undermines confidence in the robustness of advice, especially under conditions of uncertainty and environmental change (Saltelli et al. 2020). Management Strategy Evaluation (MSE) offers a structured approach to address these limitations, by testing whether the current assessment routine is robust to plausible scenarios and uncertainty. MSE also provides a framework to introduce ecosystem information and test ecosystem-informed hypotheses (Moor 2023) and make progress towards the explicit consideration of ecosystem dynamics (Trenkel et al. 2023).

Currently in the Northeast Atlantic, management strategies for forage fish and other commercial fisheries focus primarily on achieving the Maximum Sustainable Yield (MSY), driven by policy mandates (e.g., EU Common Fisheries Policy, UK Fisheries Act 2020), while adhering to the Precautionary Approach (PA; (Hilborn et al. 2020; Walters et al. 2005)). In the case of NEA mackerel, ICES recommends Total Allowable Catch (TAC) advice based on quantitative single-species stock assessments to estimate fishing mortality (F) and Spawning Stock Biomass (SSB), and then set a target fishing mortality (F_{MSY}) and a biomass threshold (MSY $B_{trigger}$), which, if biomass falls below, fishing pressure is reduced to limit the risk of the stock falling below a limit reference point, B_{lim}. However, ensuring healthy fish stocks requires more than just achieving MSY (ICES 2023b). The demographic composition of fish populations, including age and size structure, is crucial for sustainability and resilience (Griffiths et al. 2024).

While fishing may affect the availability of forage fish for predators (e.g., Frederiksen et al. 2004), forage fish recruitment is also highly variable in response to environmental conditions (e.g., temperature, Lindegren et al. 2018) and the availability of zooplankton prey (e.g., Clausen et al. 2018). In 2023, the European Union (EU) and the United Kingdom (UK) jointly requested evidence from ICES to better understand how ecosystem considerations were integrated into single-stock advice for forage fish species (ICES 2023a). There are divergent views about whether current advice adequately accounts for predator needs, as reflected in the recent arbitration between the UK and EU (Permanent Court of Arbitration 2025) in respect to the prohibition of industrial sandeel fishing in UK waters of the North Sea: one side of the argument is that predator needs are taken into account by assessments in which variable predation mortality is included, by varying natural mortality, to estimate surplus biomass available for commercial harvest. The other side of the argument is that predator needs are not accounted for, as single species assessments prioritise keeping target species at a level which enables successful recruitment, and thus single-species advice is provided without investigating how alternate fishing strategies impact predators or what the required prey biomass might be to support current predator levels or their recovery. Including predation information as variable natural mortality in stock assessments is a step towards EBFM. However, the priority of management remains MSY; the needs of predators are not explicitly considered (ICES 2013, 2023a).

Opportunities for Ecosystem-Informed Advice

MSE and strategic ecosystem models offer frameworks for assessing the potential impacts of alternative management strategies on both forage fish populations and their predators, facilitating informed decision-making and promoting ecosystem resilience (Chagaris et al. 2020; Kell et al. 2024; Craig and Link 2023; M. C. Siple, Essington, and Cope 2021; Moor 2023; Hill et al. 2020; Fulton 2022; Spence et al. 2020). MSE provides a framework for evaluating ecosystem-informed fisheries advice for single-species and is applicable to data-rich and data-limited stocks (Fischer, 2026; Perryman, Kerr, and Melnychuk 2021; Kaplan et al. 2021; Punt et al. 2016). By conditioning an Operating Model (OM) that represents resource dynamics on ecosystem processes, or by using a multi-species Operating Model (e.g., Mackinson et al. 2018), MSE enables the development of ecosystem-informed advice by acknowledging multi-species interaction and ecosystem indicators which could be embedded using operational objectives, such as Ecological Reference Points, linked to management objectives (Plagányi et al. 2014).

The approach enables decision makers and stakeholders to evaluate, prior to implementation, the trade-offs between conflicting objectives such as maximising catch while maintaining resources above levels that impair productivity. For example, Kell et al. (2024) demonstrated how MSE can integrate information from multiple sources to address both fishery and ecological objectives for a data-poor small pelagic stock. Ecosystem modelling and life history theory informed the OM, and management procedures based on empirical indicators were tested for robustness across scenarios with variable natural mortality and recruitment (Moosa and Butterworth 2024; Schiano et al. 2024). By incorporating key ecosystem interactions, such as predator-prey dynamics, and variability into OMs, MSE provides a quantitative tool to assess the robustness of harvest control rules (HCRs). Using MSE to stress-test existing management frameworks, allows the robustness of existing management frameworks to be evaluated considering ecosystem structure and climate uncertainty, and to guide adaptive management strategies that align with broader ecosystem objectives (Punt et al. 2016; Peterson et al. 2025).

One-way coupling of single-species MSE with strategic ecosystem models allows evaluation of the robustness of management strategies against historical conditions, including environmental variability and changes in stock productivity. However, structural inertia within management systems and regulatory constraints have impeded progress toward achieving ecosystem objectives (Christie 2005; Marshak et al. 2017; Craig and Link 2023). Yet implementing EBFM is not only a scientific and social-ecological imperative but also a legal obligation in the Northeast Atlantic (e.g., European Parliament and Council 2008; UK Government 2020). Decision-makers are therefore increasingly acknowledging the necessity of adopting ecosystem-informed approaches to assess risks, trade-offs, and broader impacts associated with alternative fisheries management strategies (e.g., ICES 2023c). Pragmatic solutions for integrating ecosystem information into decision-making processes are being proposed, including co-creating decision support tools with stakeholders and developing ERPs tailored to specific ecological needs (e.g., Howell et al. 2021; Bentley et al. 2021; Chagaris et al. 2020). While complex models come with caveats, the risk of inaction far exceeds the risk of imperfect action given the rapid deterioration of marine ecosystems under cumulative environmental and anthropogenic pressures (Fulton 2021; Pikitch et al. 2004; Free et al. 2020).

Despite repeated scientific warnings and the loss of Marine Stewardship Council (MSC) certification, catch levels for NEA mackerel have routinely exceeded recommended limits since 2009, with individual coastal states setting unilateral quotas in the absence of binding international agreements. This persistent over-exploitation has contributed to a marked decline in stock biomass since 2015, raising concerns for both the fishery's sustainability and the broader marine food web, as mackerel is a vital prey resource for seabirds, marine mammals, and larger fish. The ongoing failure to achieve consensus on quota sharing among all fishing nations further underscores the limitations of traditional single-species management in addressing transboundary, climate-impacted stocks. These challenges highlight the urgent need for ecosystem-informed, adaptive management approaches that can account for environmental variability, ecological interactions, and the socio-political realities of international fisheries governance.

In this paper, we use a strategic Management Strategy Evaluation (MSE) to stress-test a generic ICES-style advice rule for Northeast Atlantic mackerel, rather than to provide up-to-date tactical advice for current quota setting. By stress-test we mean a historical closed-loop backtest of the advice framework already in use: a single management procedure is run over 2005–2020 under alternative Operating Model hypotheses about productivity and predation. We do not compare candidate harvest strategies for hypothetical futures; we ask whether the current ICES-style rule, if fully implemented, would have delivered conventional and ecosystem objectives. The Operating Model is conditioned on a fixed ICES assessment configuration and associated reference points; subsequent benchmarks that revise stock dynamics or reference points for tactical advice are not required for this purpose. Our focus is on the robustness of the harvest control rule structure—an F_{MSY}-based target with an MSY $B_{trigger}$ threshold and TAC stability constraints—to plausible uncertainty in natural mortality, recruitment dynamics, and ecological objectives. In particular, we ask whether such an advice rule, implemented as a full closed-loop management procedure over the historical period 2005–2020, would have been sufficient to maintain both conventional single-species objectives and additional ecosystem-oriented objectives, such as forage availability, age structure, and rebuilding time, under alternative hypotheses about productivity and predation mortality.

Methods

The Management Strategy Evaluation (MSE) stress-tests a single, existing management procedure—the ICES advice rule (ICES-AR)—rather than comparing alternative candidate procedures. The analysis is strategic rather than tactical, implemented as a backtest over the historical period as it is not intended to update stock status or provide advice on quotas (Bentley et al. 2020). Age-structured Operating Models conditioned on alternative natural-mortality and recruitment hypotheses are described below (Figure 1); the closed-loop backtest, harvest control rule, and performance metrics are in Sections 2.3–2.4. Operating Model equations and software details are in the Supplementary Material; code and data are at

https://github.com/flrpapers/erp_nea_mackerel.

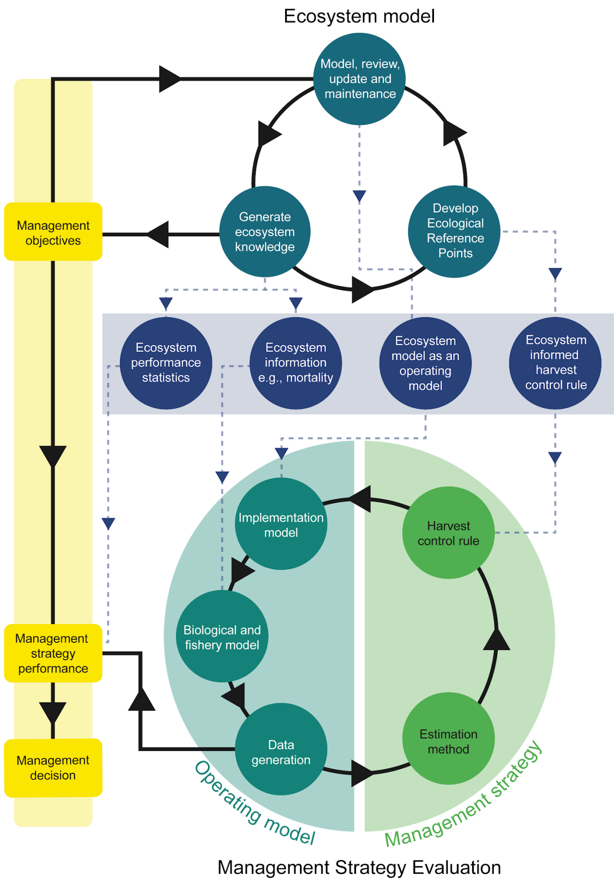

Figure 1. Framework for Management Strategy Evaluation showing conditioning of the operating model on ecosystem data and knowledge based on a single-species assessment, management procedure, and feedback loops.

Operating model

The Operating Model is an age-structured population model (ages 0–12) conditioned on the ICES State-space Assessment Model (SAM) for NEA mackerel (WGWIDE inputs, 2022 assessment configuration; years 1980–2020). We retain this configuration deliberately: the study evaluates strategic options for the advice rule and Ecological Reference Points, not tactical stock status or quota advice, so updating to a later benchmark is not required. For each natural-mortality hypothesis, the assessment was refitted to the same catch and abundance indices with scenario-specific M-at-age; stock numbers, fishing mortality, and biological parameters were taken from that fit. Initial-condition uncertainty was propagated with Monte Carlo iterations drawn from the SAM fit (`SAM_uncertainty`; Supplementary Material). Mean fishing mortality is summarised over fully selected and mature ages 4–8.

Ecosystem information enters the Operating Model by one-way coupling to strategic Ecopath with Ecosim (EwE) models (Moor 2023). Relative predation-mortality (M_2) trends for mackerel were extracted from models where such estimates were available. These included models of the North Sea (Mackinson et al. 2018), Norwegian/Barents Sea (Bentley, Serpetti, and Heymans 2017), and West Coast of Scotland (Serpetti et al. 2017) models (Figure 2). None of the EwE models spans the full stock range; they supply plausible temporal M variability rather than stock-specific rates. For integration into ICES advice, such models and their outputs would need to be reviewed by independent reviewers under the Working Group for Multispecies Assessment Methods (WGSAM; ICES, 2019). While the North Sea model has been through this review process, the other published models used in this study have not. But, for the purpose of this paper, we consider these trends as though they are equally plausible for the benefit of the retrospective stress test and to demonstrate a route for integrating ecosystem information into MSE.

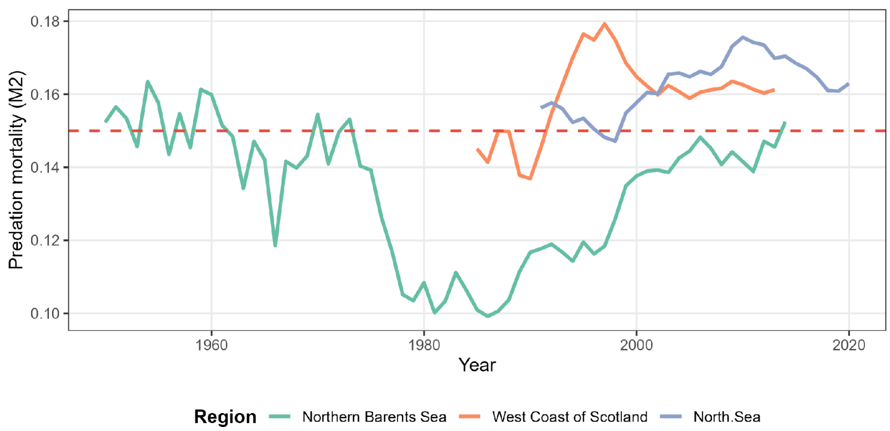

Figure 2. Relative predation mortality (M_2) trends from Ecopath with Ecosim models for the Northern Barents Sea, West Coast of Scotland, and North Sea, scaled relative to the ICES baseline $M = 0.15$ (dashed red line). These series inform the time-varying natural mortality scenarios in the operating model; the West Coast of Scotland series is shown for context but was not used as a separate Operating Model (see text).

Age-specific natural mortality follows the Lorenzen mass-at-age relationship (Lorenzen 2022) in **all** Operating Model scenarios:

$$M_{a,y} = M_{ref,y}\left(\frac{W_{a,y}}{W_{ref,y}}\right)^{-0.288},$$

where $W_{a,y}$ is stock weight-at-age and W_{ref} is the reference weight at age at 50% maturity. Four natural-mortality hypotheses are used. Two are independent of the EwE series: a *constant* $M = 0.15$ at all ages, matching the ICES assessment assumption; and a *Reference* case in which $M_{a,y}$ follows the Lorenzen mass-at-age schedule (Eq. [eq:lorenzen]) scaled so that M at the reference weight equals 0.15, so younger ages have higher M and older ages lower M as weights change over time. Two further hypotheses perturb the Reference schedule: an *Environmental Driver* case, in which M is rescaled so that it tracks the Norwegian/Barents Sea EwE M_2 series (extended to 2020 by holding the final ratio for the recent years) while relative M across ages is preserved; and a *High M* case, in which

Reference M is increased by 25%. The North Sea and West Coast of Scotland EwE series (Figure 2) are shown for context but were not used to condition Operating Models. Total M is partitioned as $M = M_1 + M_2$ with background mortality $M_1 = 0.025\,y^{-1}$ and predation mortality $M_2 = M - M_1$.

The ICES assessment assumes constant $M = 0.15$ and does not fit a stock–recruitment relationship (SRR) internally (SRR parameters are estimated post hoc from SSB and recruitment). The Operating Model set (Table 1) crosses these natural-mortality hypotheses with three Beverton–Holt SRR specifications: a FishLife prior on steepness ($SRR.i$), $h = 0.9$ ($SRR.ii$), and depensation at low SSB ($SRR.iii$). Six scenarios are presented (the reference case and the main effects): namely the four natural-mortality hypotheses and the prior-steepness SRR, plus $h = 0.9$ and depensation variants of the Reference case.

Operating-model structures used in the MSE (each run under both recruitment processes in Section 2.3). All M scenarios use the Lorenzen mass-at-age schedule (Eq. [eq:lorenzen]); they differ only in the scaling of $M_{ref,y}$.

Scenario	**Natural mortality (M_{ref} scaling)**	**Stock–recruitment**
Reference	Reference ($M = 0.15$ at reference weight)	SRR.i (prior h)
Robustness 1	Constant ($M = 0.15$ ICES assumption)	SRR.i (prior h)
Robustness 2	Environmental Driver (EwE M_2)	SRR.i (prior h)
Robustness 3	High M (Reference $\times\ 1.25$)	SRR.i (prior h)
Robustness 4	Reference M	SRR.ii ($h = 0.9$)
Robustness 5	Reference M	SRR.iii (depensation)

Management Procedure (ICES advice rule)

Control parameters for the ICES advice rule (Figure 3) were taken from the 2021 reference points used with the conditioning assessment: $F_{MSY} = 0.26$, MSY$B_{trigger} = 2\,580\,000$ t, and $B_{lim} = 2\,000\,000$ t. Each year the management procedure obtains assessed SSB ($\hat{SSB}_y$) from a SAM fit (Section 2.3). The HCR sets

$$F_y = \{F_{MSY}\ if\ \hat{SSB}_y \geq MSYB_{trigger},\ F_{MSY}\frac{\hat{SSB}_y}{MSYB_{trigger}}\ if\ B_{lim} < \hat{SSB}_y < MSYB_{trigger},\ F_{min}\ if\ \hat{SSB}_y \leq B_{lim},$$

with $F_{min} = 0.005$. The corresponding TAC is obtained from the projected catch at F_y. When $\hat{SSB}_y > MSYB_{trigger}$, interannual TAC change is constrained to the interval $[0.8, 1.25]$ times the previous TAC (-20% to $+25\%$); the bound is not applied below the trigger. Advice is assumed fully implemented (catch equals TAC). No Ecosystem Control Points (ECPs) enter the HCR: ecological quantities are evaluated only as performance metrics (Section 2.4).

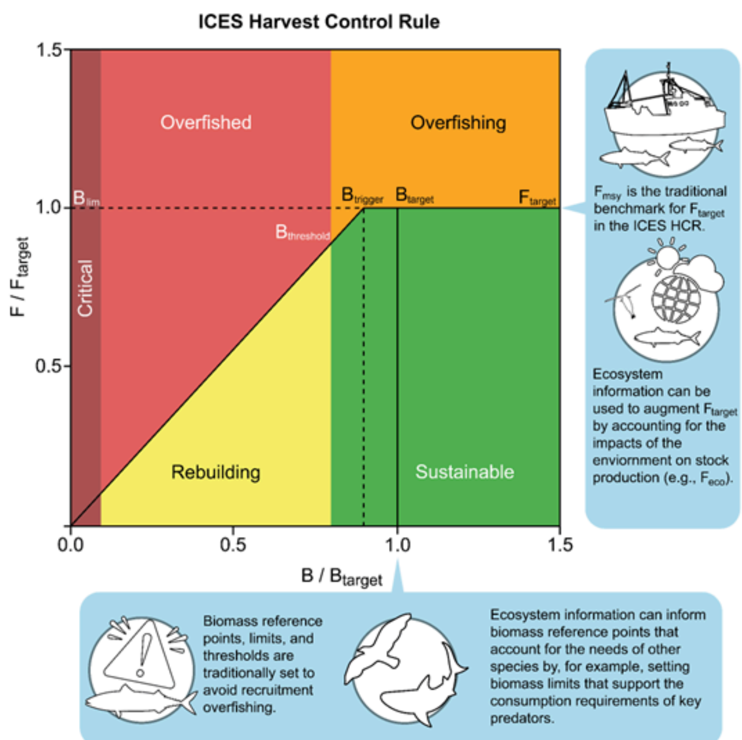

Figure 3. Harvest Control Rule (HCR) showing target fishing mortality (F_{MSY}), biomass threshold ($MSYB_{trigger}$), and limit (B_{lim}).

Backtest

The closed-loop backtest runs from 2005 to 2020; with retrospective peels. Simulations were conducted using the FLR framework (L. T. Kell et al. 2007). We implement a full closed-loop management procedure with annual re-estimation of stock status (Punt et al. 2016; ICES 2020). In each simulated year y the procedure: (i) forms a survey index proportional to true Operating Model stock numbers at age, with independent lognormal observation error (CV ≈ 0.1); (ii) constructs a stock assessment data set that uses catch-at-age from the Operating Model with constant $M = 0.15$; (iii) fits SAM to that stock and the noisy indices; (iv) applies Eq. [eq:hcr] to assessed SSB to set the TAC; and (v) implements that TAC exactly in the Operating Model (no implementation error). The estimation model keeps fixed $M = 0.15$ while Operating Models use mass-at-age schedules with alternative M_{ref} scaling, the procedure does not "know" the true Operating Model M—as required when alternative natural-mortality hypotheses are tested (ICES 2020).

Each of the six Operating Model scenarios (Table 1) was run under two assumed recruitment processes (twelve configurations in total; $n = 100$ iterations each):

1. **Epistemic:** autocorrelated (AR1) recruitment deviations superimposed on the smoothed historical trend from the conditioning assessment (an explainable, potentially reducible directional signal).

2. **Aleatory:** the same autocorrelated (AR1) deviations without the trend (irreducible process noise; coefficients in the Supplementary Material).

Performance metrics

We evaluate the advice rule with two traditional single-species metrics (status and yield) and three Ecological Reference Points (ERPs: forage, ABI_{MSY}, and rebuilding time; Table 2). ERPs are used here as performance metrics computed from the Operating Model; in contrast Ecosystem Control Points (ECPs) would adjust the HCR.

Performance metrics used to evaluate the ICES advice rule

Metric	Definition	Reported in
Status	$P\left(SSB/B_{MSY} \geq 1\right)$; $P\left(F/F_{MSY} \leq 1\right)$	Figs 6, 7
Yield	Catch relative to MSY (pretty-good yield near $0.80 \times MSY$)	Fig 7
ABI_{MSY}	Ratio of old-age number share to that at F_{MSY} equilibrium (Griffiths et al. 2024)	Figs 5, 7
Forage	Predation removals (Eq. [eq:forage]), relative to MSY-equilibrium forage	Figs 5, 7
Rebuilding	Years to restore the threshold under reduced F (target: $P\left(T_{rebuild} \leq 0\right)$; limit: $P\left(T_{rebuild}/8.5 \leq 1\right)$, one generation $= 8.5$ y); negative if already above target	Figs 5, 7

Forage availability is the biomass removed by predation, analogous to fishery yield (Kell et al. 2024):

$$Forage_y = \sum_a W_{a,y} N_{a,y} \frac{M_{2,a,y}}{Z_{a,y}} \left(1 - e^{-Z_{a,y}}\right),$$

with $Z_{a,y} = F_{a,y} + M_{a,y}$. No forage limit or target enters the HCR; results report forage relative to the MSY-equilibrium forage level.

ABI_{MSY} (Griffiths et al. 2024) is the ratio of the proportion of numbers at or above a reference old-age threshold in the simulated stock to the same proportion at F_{MSY} equilibrium (threshold age = first age at which cumulative numbers reach 90% of the F_{MSY} age distribution, excluding age 0).

Rebuilding time is the number of years to restore a chosen threshold under a rebuilding plan (reduced fishing mortality); negative values mean the stock is already above the threshold. The rebuilding limit expresses this relative to one generation (8.5 years). It allows a comparison of recovery speed across OMs and objectives.

Results

Results are summarised in three ways. Figure 4 shows trajectories (medians and percentile ribbons). Figure 5 reports Kobe-style joint status probabilities for F and SSB. Figure 6 summarises outcomes within three contiguous backtest blocks—start (2010–2014), transition (2015–2019), and end/gain (2020)—as traffic lights. Colour coding follows a target-versus-limit logic (Table 3). **Targets** ask whether performance is acceptably close to the desired level and use a three-colour scale: red (poor), amber (marginal), green (acceptable). For probability-based targets the amber/green cut-points are either $0.50/0.66$ (SSB, ABI, forage) or the more precautionary $0.33/0.50$ (F, rebuilding already above threshold); yield uses median catch/MSY with pretty-good-yield bands at 0.80 and 1. **Limits** ask whether a more severe threshold is avoided with high probability and are binary (red/green only): rebuilding within one generation must succeed with $P \geq 0.95$; ABI and forage at half the MSY-equilibrium level must succeed with $P \geq 0.05$ (failure rarer than 5%).

Traffic-light quantities and colour breakpoints used in Figure 6. Target panels use a three-colour scale; limit panels use a binary scale with a high (rebuilding) or low-failure (ABI, forage) probability threshold.

Type	Panel	Quantity	Red	Amber	Green
Target	SSB	$P\big(SSB/B_{MS\}$	< 0.50	$0.50–0.66$	≥ 0.66
	F	$P\big(F/F_{MSY} \leq$	< 0.33	$0.33–0.50$	≥ 0.50

Type	Panel	Quantity	Red	Amber	Green
	Yield	median(catch/MSY)	< 0.80	0.80–1.00	≥ 1
	ABI	$P(ABI \geq 1)$	< 0.50	0.50–0.66	≥ 0.66
	Forage	$P(Forage \geq$	< 0.50	0.50–0.66	≥ 0.66
	Rebuild	$P(Rebuild$:	< 0.33	0.33–0.50	≥ 0.50
Limit	ABI	$P(ABI \geq 0.$	< 0.05	—	≥ 0.05
	Forage	$P(Forage \geq$	< 0.05	—	≥ 0.05
	Rebuild	$P(Rebuild$: gen.)	< 0.95	—	≥ 0.95

Reference Case

Time series of recruitment, fishing mortality, SSB, catch, forage, and ABI (relative to MSY benchmarks) from the reference case under the two recruitment scenarios are shown in Figure 4. The pink ribbons span the 50% (25th–75th) and 95% (2.5th–97.5th) percentile intervals, the black line is the median and the grey "worm" shows a single Monte Carlo simulation. The horizontal reference lines are the MSY target (grey), limit (red), and virgin (green) levels.

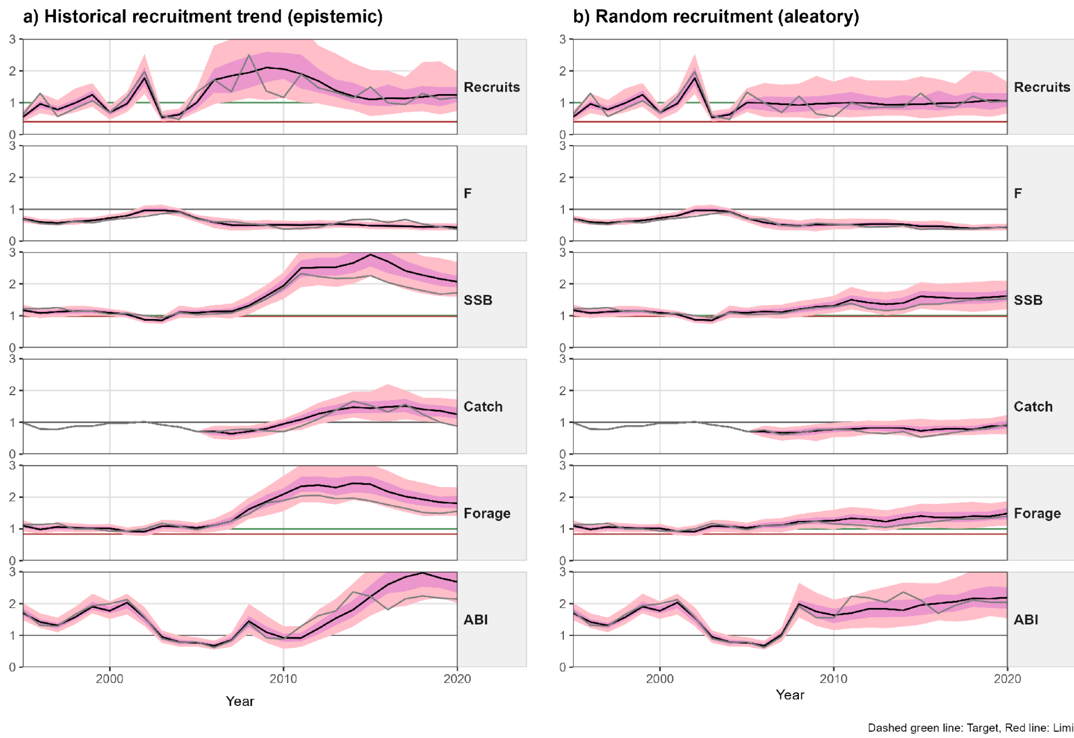

Figure 4. Time series for base case of recruitment (log scale), fishing mortality, Spawning Stock Biomass (SSB), catch, forage, and Age-Based Indicator (ABI) for the reference operating model under (a) epistemic (historical trend) and (b) aleatory (random) recruitment scenarios, 2005–2020. Grey line: single simulation; pink ribbons: 50% (25th–75th) and 95% (2.5th–97.5th) percentile intervals; black line: median; horizontal lines: virgin (green), limit (red), and MSY target (grey).

Under historical (epistemic) recruitment trend, where an increase was seen in 2005, then a decline after 2010, SSB initially rose above B_{MSY} and fishing pressure fell. After about 2015, the TAC constraint prevented rapid catch reductions as biomass declined, pushing F above F_{MSY}; by 2020 SSB had returned toward B_{MSY}. ABI remained above the MSY level. Outcomes therefore depended on the recruitment pattern as well as on the HCR.

Under random (aleatory) recruitment, the HCR recovered SSB and catches toward MSY levels. F remained consistently below F_{MSY} (median ~0.8), with some cycling due to the TAC constraints, while catch and SSB uncertainty widened as recruitment variability accumulated. Forage stayed well above the MSY-equilibrium reference level; ABI increased as strong year classes entered the older ages.

Robustness Trials

Robustness is evaluated across the Operating Model scenarios (Table 1) in Figure 5. Stock status relative to MSY is summarised with Kobe-style phase probabilities (Kell, Kimoto, and Kitakado 2016): $F > F_{MSY}$ & $SSB < B_{MSY}$ (red), $F < F_{MSY}$ & $SSB > B_{MSY}$ (green), $F > F_{MSY}$

& $SSB > B_{MSY}$ (orange), and $F < F_{MSY}$ & $SSB < B_{MSY}$ (yellow). Under high steepness the stock is mostly in the green quadrant under both recruitment regimes. Under aleatory recruitment, higher M—especially the trend from the EwE model—increased the probability of being in the red-quadrant. Under the epistemic recruitment trend, the reduction in recruitment increased the probability of overfishing but not being overfished (orange). TAC constraints, limiting catch reductions, increased the probability of overfishing and being overfished (red).

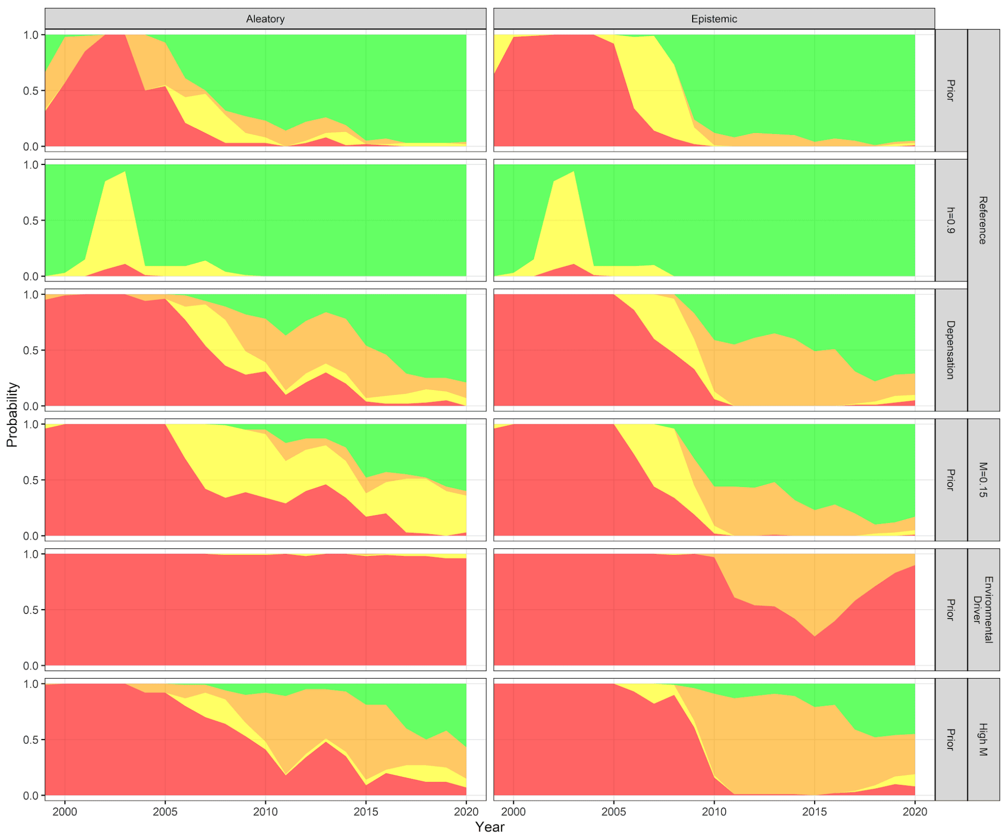

Figure 5. Probabilities of stock status (SSB/B_{MSY} vs. F/F_{MSY}) for the Reference Case and Robustness trials (rows; aligned with Table 1) under aleatory (random) and epistemic (historical) recruitment (columns), 2005–2020. Colours: $F > F_{MSY}$ & $SSB < B_{MSY}$ (red), $F < F_{MSY}$ & $SSB > B_{MSY}$ (green), $F > F_{MSY}$ & $SSB > B_{MSY}$ (orange), and $F < F_{MSY}$ & $SSB < B_{MSY}$ (yellow).

Summary of Performance

The metrics in Tables 2–3 are summarised in Figure 6 as traffic lights for the different backtest periods; start (2010–2014), transition (2015–2019), and end/gain (2020). Target panels use the three-colour scale; and limit are binary (Table 3). Panels show epistemic (a) and aleatory recruitment (b); rows correspond to Operating Model and columns to time periods. Under aleatory recruitment, SSB and F were mostly near targets and pretty-good yield was mostly achieved; rebuilding times were short or negative, ABI recovered toward or above ABI_{MSY}, and forage tracked SSB near the MSY-equilibrium level. Performance deteriorated mainly under environmental M trend.

Under the epistemic trend in recruitment, SSB rose then fell toward B_{MSY}, with F able to exceed F_{MSY} late in the backtest because of TAC constraints; following advice implied lower catches early and higher catches in the transition and end periods relative to history. Performance deteriorated mainly under elevated M or depensatory recruitment environmental M trend.

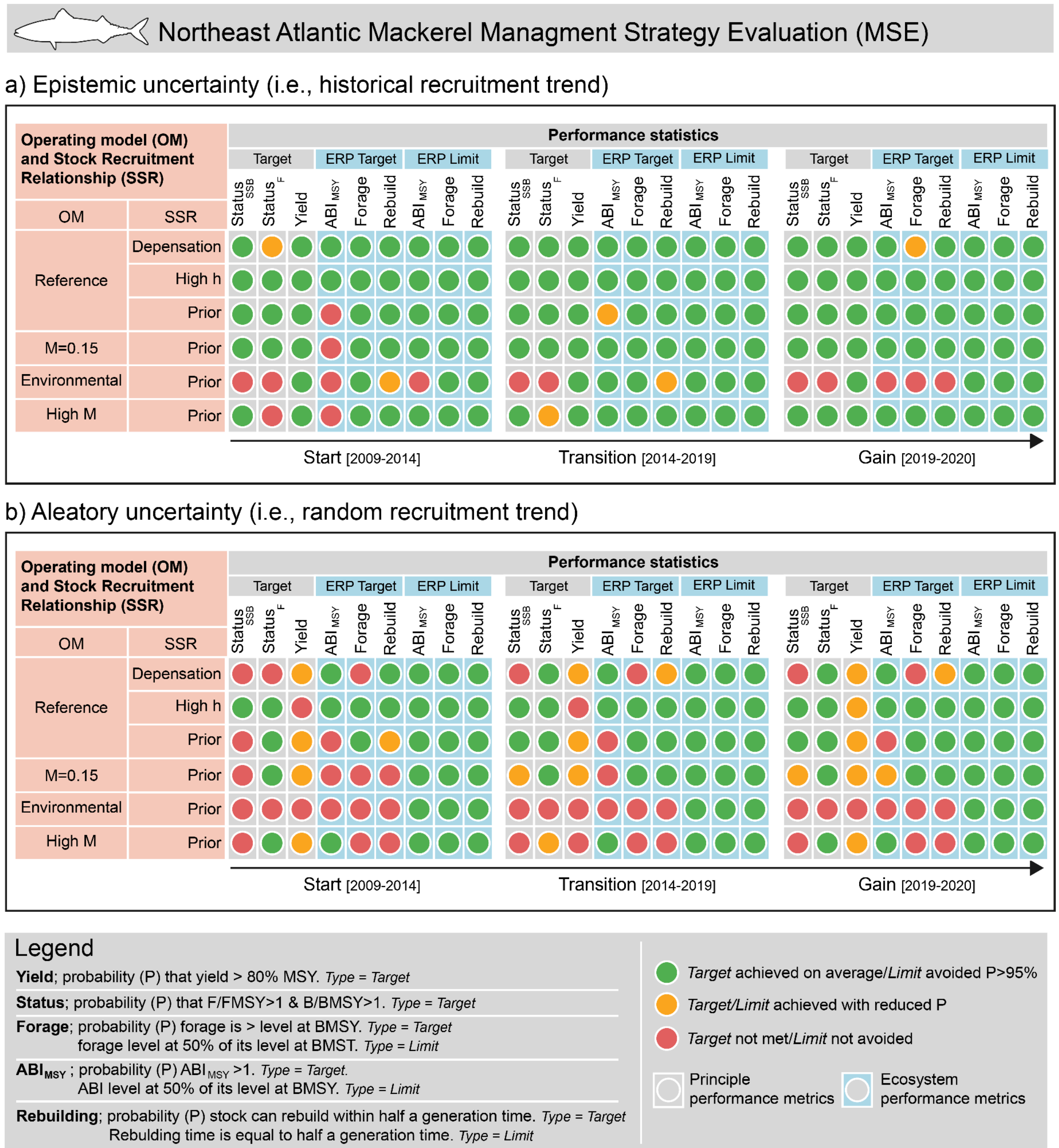

Figure 6. Traffic-light summary of traditional targets and ERP targets/limits across start (2010–2014), transition (2015–2019), and end/gain (2020) periods and operating model scenarios under (a) epistemic and (b) aleatory recruitment. Target panels use a three-colour scale; limit panels are binary with high (or low-failure) probability thresholds (Table 3).

For the reference Operating Model under aleatory recruitment, target lights for SSB, F, and yield were mostly green through the transition and end periods; forage stayed above its limit and near but below its target; ABI improved; rebuilding target and limit lights indicated little need for a rebuilding response. Under epistemic recruitment, lower recruitment rather than fishing alone pulled stock and yield below MSY targets late in the backtest, although the HCR did reduce catch.

Across robustness trials under aleatory recruitment, high steepness kept status and ERP targets largely green. Depensation lengthened median rebuilding times beyond one generation and reduced forage relative to the reference case, with rebuild-limit lights more often red. Elevated M, especially the EwE trend, increased overfishing probability and lowered forage, with SSB more often near or below B_{MSY}, while catch often remained in the pretty-good-yield range.

Under epistemic recruitment, several scenarios showed red or amber status and yield targets as recruitment declined after 2010 and TAC constraints limited catch reductions, so F exceeded F_{MSY}. Combined elevated M and depensation increased red lights for ERP targets and limits: forage often fell below the MSY-equilibrium target and toward or below the forage limit (half that level), rebuilding exceeded one generation, and ABI deteriorated even when SSB remained near B_{MSY}.

Discussion

A historical closed-loop backtest within MSE asks whether an existing advice rule remains fit for purpose under single-species and ecosystem objectives. For the ICES advice rule, ERPs can be scored alongside conventional status and yield without changing the HCR. Here we shown that under the assumption of aleatory recruitment the rule largely met single-species targets and ERP targets/limits; under epistemic recruitment decline scenario, elevated M, or depensation it did not. That contrast implies that the level of precaution needed depends on whether uncertainty is irreducible noise or a directional change in productivity. Explicit Ecosystem Control Points in alternative HCRs remain a logical next step; here ERPs are performance metrics only.

Earlier ICES evaluations of the advice rule did not fully explore epistemic versus aleatory recruitment, nor population structure and predator needs. In this case study the rule mostly achieved both conventional and additional ecosystem objectives unless M was elevated or trended (Environmental Driver / High M scenarios informed by EwE M_2 series). Recommended F can then fall out of step with ecosystem state if the assessment retains constant M. Many ICES assessments already take multispecies M from the Working Group on Multispecies Assessment Methods (WGSAM); the assessment in the management procedure used here assumed constant $M = 0.15$. Keeping that assumption in the estimation model while varying Operating Model M is what allows the closed-loop backtest to model the mismatch (ICES 2020; Punt et al. 2016).

A related limitation of conventional advice is the narrow focus of B_{lim} on recruitment impairment. A fixed limit does not address rebuilding timelines, age structure, or prey for predators. That is why we scored ABI_{MSY}, rebuilding time, and forage as complementary performance metrics, with traffic lights distinguishing targets (proximity to desired levels) from limits (high-probability avoidance of severe thresholds; Table 3; Figure 6). Rebuilding time is an ERP comparator of recovery speed, not a B_{MSY} status probability: short or negative values under most aleatory scenarios indicate transient drops in SSB whereas generation-scale delays under depensation and epistemic recruitment decline flag when a rebuilding response would be needed—including when environmental change rather than fishing alone drives the breach. Thresholds above the limit, including forage or age-structure triggers, allow that evaluation before impairment.

These findings align with Bastardie et al. (2022): advice frameworks may cope with short-term variability yet struggle with longer directional change. Empirical risk-equivalent approaches (Duplisea et al. 2021) and ecosystem-informed fishing mortality (F_{eco}; (Bentley et al. 2021)) offer routes to embed such considerations in advice. Our results remain strategic guidance on HCR and ERP design, not tactical advice for the current NEA mackerel stock. Conditioning on the latest assessment would shift the exercise toward updating status rather than testing strategy. Re-running the workflow on updated Operating Models is a natural extension when operationalising ERPs, but is not required to show that ERPs can be evaluated alongside an existing advice rule.

Next steps

Feasibility does not require fully coupled ecosystem models. As Moor (2023) argued, tactical EBFM can start with simplified models, expert judgement, and strategic ecosystem information once objectives and risk levels are clear. A practical sequence is to (i) report ERPs alongside single-species metrics in advice products; (ii) agree with managers and stakeholders which could become Ecosystem Control Points—including what “predator needs” means for forage (ICES 2023a; M. C. Siple, Essington, and Cope 2021; Chagaris et al. 2020); and (iii) simulation-test HCR adjustments (for example reducing F when forage falls below a pre-agreed threshold). The one-generation rebuilding limit used here is already a policy-style choice of recovery horizon; rebuild-based reference points ($T_{rebuild}$, $B_{rebuild}$) offer a related generalisation for later work (Kell et al. submitted). Metrics still need verification and validation (Saltelli et al. 2020), and single-species models omit interactions that prevent simultaneous multi-species MSY optimisation (Del Santo O’Neill et al. 2024; Spence et al. 2024).

Figure 6 is intended as a communication format as well as a scientific summary: target versus limit lights across periods make trade-offs visible. ICES abandoned traffic lights on advice sheets in 2021; the device here is a stakeholder dashboard, not a return to that convention. Early engagement helps frame actionable EBFM requests (Dickey-Collas et al. 2014; Cvitanovic, McDonald, and Hobday 2016); contested quota sharing, loss of MSC certification (Marine Stewardship Council (MSC) 2023), and disagreement over ecosystem information in forage-fish advice (ICES 2025) show the cost of weak shared interpretation. ICES engagement routes and participatory modelling can co-develop ERPs and risk tolerances (ICES 2023d; Stephenson et al. 2019; Deith et al. 2021; Howell et al. 2021); backtests are useful in that setting because consequences under known history are concrete (Kell, Kimoto, and Kitakado 2016; Kell et al. 2021). However, it is important to recognise that ICES advice is not management, and thus a governance gap exists between identifying ecosystem trade-offs and making the often contested decisions needed to resolve them; operational EBFM therefore requires managers to specify objectives and risk tolerances.

Conclusions

EBFM can advance through incremental stress-testing of current advice. For NEA mackerel, the ICES advice rule was broadly effective under aleatory recruitment, but became misaligned with single-species and ecological objectives when M was elevated or recruitment declined systematically; TAC constraints amplified late overfishing under epistemic recruitment decline. ABI, rebuilding time, and forage, summarised with target and limit traffic lights, make those failures visible alongside conventional metrics.

Backtesting grounds evaluation in observed dynamics and supports stakeholder discussion of trade-offs. Ecosystem information can enter advice without fully coupled models when strategic ecosystem inputs, clear objectives, and agreed risk levels are available. Moving from ERPs as performance metrics to ECPs in HCRs requires institutional commitment and inclusive deliberation as much as further technical work.

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Data Availability Statement

The code for conducting the MSE is available at https://github.com/flrpapers/erp_nea_mackerel, and the data, inputs and results from the Dryad Digital Repository https://doi.org/10.5061/dryad.98sf7m0wk

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